

Cognitive Science — Module 04: Cognition — Study Questions

1. What is variational free energy (VFE)? Explain it as a "badness score" for your model of the world.
2. What are the two components of free energy minimization (accuracy and simplicity)? Why does the brain need both?
3. Using the conspiracy theory example, explain how the accuracy-complexity trade-off works. What goes wrong when someone adopts a conspiracy theory?
4. What is precision in Active Inference? Describe it using the "volume knob" metaphor. Give an example of turning precision up and down.
5. How does attention relate to precision weighting? What does it mean to "turn up the volume" on visual processing when you're reading?
6. What neurotransmitters modulate precision? What does acetylcholine do? What about dopamine? How do these relate to learning?
7. How might ADHD be understood as a precision-weighting disorder? What would it feel like if you couldn't properly adjust your attention volume knobs?
8. Explain the "dark room" problem. Why don't agents just hide in a dark room to minimize surprise? Give all three answers.
9. How does Active Inference explain confirmation bias? Is confirmation bias always bad, or does it sometimes help?
10. Compare the Active Inference explanation of the sunk cost fallacy with the standard economics explanation. Which is more satisfying and why?
11. How does Active Inference explain delusions in schizophrenia? Why is the delusional person not being "irrational" from their brain's perspective?
12. Why does the "compassionate understanding" framing matter clinically? How does viewing psychiatric symptoms as rational consequences of precision settings change treatment approaches?
13. What is epistemic value? How does it explain why humans are naturally curious — why we explore, read, and ask questions?
14. What is expected free energy? How does it combine the desire for reward (pragmatic value) with the desire for knowledge (epistemic value)?

15. Critically evaluate: If the brain truly minimizes free energy, why do people sometimes seek out horrifying experiences (horror movies, skydiving)? Does this challenge or support Active Inference?
16. What is "metacognition" in Active Inference terms? When you say "I'm not sure about my answer," what level of the generative model is generating that uncertainty assessment? How does this differ from first-order uncertainty about the world?
17. How does Active Inference explain the experience of "flow" — the state of effortless, absorbed performance that athletes and musicians describe? What is happening to prediction errors and precision during flow, and why does it feel so rewarding?
18. Explain the "free energy" of emotion: how does Active Inference account for feelings like anxiety (high expected uncertainty), boredom (low prediction error in an unchanging environment), and curiosity (high epistemic value)? Why aren't emotions separate from inference?
19. What is the relationship between sleep and free energy minimization? During sleep, the brain cannot act on the world or receive new observations. What computational work is the sleeping brain doing, and why is it essential for maintaining a good generative model?
20. How does the precision-weighting framework explain why stress impairs cognitive performance? When cortisol rises, what happens to the balance between prior beliefs and sensory evidence, and how does this create the subjective experience of "not thinking clearly"?